

RAGLEN SYSTEM BALANCE, INC.

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To JESSEE HEATING AND AIR CONDITIONING

3025 SOUTHGATE LN.

CHICO, CA 95928

LETTER OF TRANSMITTAL

Date
NOVEMBER 20, 2023

Job No.
20531B

Attention
BRYAN GRAY

RE:
NANOTECH ENERGY FUME HOOD SURVEY

WE ARE SENDING YOU ☐ Attached ☐ Under separate cover via _____ the following items:

☐ Shop drawings ☐ Prints ☐ Plans ☐ Samples ☐ Specifications

☐ Copy of letter ☐ Change order ☒ TEST AND BALANCE REPORT

COPIES	DATE	NO.	DESCRIPTION
1	11/20/2023		TEST AND BALANCE REPORTS FOR THE ABOVE REFERENCED PROJECT

THESE ARE TRANSMITTED as checked below:

☒ For approval ☐ Approved as submitted ☐ Resubmit _____ copies for approval

☐ For your use ☐ Approved as noted ☐ Submit _____ copies for distribution

☒ As requested ☐ Returned for corrections ☐ Return _____ corrected prints

☐ For review and comment ☐ _____

☐ FOR BIDS DUE _____ 20 _____ ☐ PRINTS RETURNED AFTER LOAN TO US

REMARKS

EMAILED BRYANGRAY@JESSEEHEATINGANDAIR.COM ON 11/20/2023 NO HARD COPIES SENT.

COPY TO _____ SIGNED: MATT BRENNAN

If enclosures are not as noted, kindly notify us at once.

CAL. LIC. # 434901

898 MAESTRO DR, SUITE 100
RENO, NEVADA 89511
(775) 747-0100



*AIR AND HYDRONIC SYSTEM
TESTING AND BALANCING*



TEST AND BALANCE REPORT

NANOTECH ENERGY FUME HOOD SURVEY

MECHANICAL CONTRACTOR: JESSEE HEATING AND AIR CONDITIONING

CONSULTING ENGINEER: -

JOB NO: 20531B

DATE: NOVEMBER 20, 2023

TECHNICIAN: MICHAEL CONNOLLY

TBE: MATTHEW I. BRENNAN



NANOTECH ENERGY FUME HOOD SURVEY

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GENERAL NOTES

1. CORRECTIONS FOR TEMPERATURE AND ALTITUDE HAVE BEEN MADE ON ALL TEST RESULTS SHOWN IN THIS REPORT.
2. BALANCE FACTORS FOR EXHAUST SIDEWALL AND SUPPLY GRILLES WERE CALCULATED FROM THE CORE AREA AND MEASURED WITH A CALIBRATED METER, MODEL, M.F.G. (SEE INSTRUMENT CERT. OF CALIBRATION.)
3. ALL INSTRUMENTS USED FOR TESTING AND BALANCING ARE TRACEABLE BACK TO THE NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY AND ARE TESTED IN OUR OWN LAB PER AABC.
4. FOLLOWING THIS SHEET ARE:
 - B. SYMBOL SHEET
 - C. INSTRUMENT SHEETS

ACRONYM SHEET

ACH	AIR CHANGES PER HOUR	HD	HEAD
ADD	ADDRESS	HD	HOT DECK
ADJ. P.D.	ADJUSTED PITCH DIAMETER	HP	HORSEPOWER
BHP	BRAKE HORSEPOWER	IC	INCHES OF WATER COLUMN
CL	CENTER	KW	KILOWATTS
CD	CEILING DIFFUSER	LA	LEAVING AIR TEMPERATURE
CD	COLD DECK	LWT	LEAVING WATER TEMPERATURE
CFM	CUBIC FEET PER MINUTE	MBH	THOUSAND BTU PER HOUR
COMP	COMPRESSOR	MIN	MINIMUM
CONT	CONTROLLER	N/A	NOT ACCESSIBLE / NOT AVAILABLE
CORR	CORRECTION	N/T	NOT TAKEN DUE TO NO VALID LOCATION
CR	CEILING RETURN	OA	OUTSIDE AIR
DB	DRY BULB TEMPERATURE	OPEN	NO TERMINAL DEVICE INSTALLED
DD	DIRECT DRIVE	PD	PITCH DIAMETER
DIA	DIAMETER	PP	PERFORATED PLATE
DNA	DATA NOT AVAILABLE	PSI	POUNDS PER SQUARE INCH
DNL	DATA NOT LISTED	RPM	REVOLUTIONS PER MINUTE
ΔP OR DP	PRESSURE DIFFERENTIAL	SCP	STATIC CONTROL POINT
(E)	EXISTING	SF	SERVICE FACTOR
EA	ENTERING AIR TEMPERATURE	SP	STATIC PRESSURE
EA	EFFECTIVE AREA / K FACTOR	SP	SETPOINT
ESP	EXTERNAL STATIC PRESSURE	SWE	SIDE WALL EXHAUST
EWI	ENTERING WATER TEMPERATURE	SWR	SIDE WALL RETURN
FE	FLOOR EXHAUST	SWS	SIDE WALL SUPPLY
FLA	FULL LOAD AMPS	TEMP	TEMPERATURE
FPM	FEET PER MINUTE VELOCITY	TSP	TOTAL STATIC PRESSURE
FR	FLOOR RETURN	V	VOLTS
FS	FLOOR SUPPLY	VP	VELOCITY
FT	FEET	WB	WET BULB TEMPERATURE
GDR	GRILLE, DIFFUSER, REGISTER	#	NUMBER OR POUND
GPM	GALLONS PER MINUTE		

Test & Balance Instrumentation Sheet

☒ THE FOLLOWING CHECKED INSTRUMENTS WERE USED TO SUCCESSFULLY MEASURE AND SET EACH DEVICE ON THIS PROJECT.

	<u>Manufacturer</u>	<u>Model No.</u>	<u>Serial No.</u>	<u>Calibration Date</u>
☐ AIR DATA MULTIMETER	SHORTRIDGE	ADM-860C	M20253	3/20/2023
☐ FLOW HOOD	SHORTRIDGE	DIGITAL 8400		N/A
☒ FLOW MODULE	EVERGREEN	S-PVF-1	2200251A	5/4/2023
☐ CAPTURE HOOD	EVERGREEN	CH-15D	2200137	N/A
☐ ROTATING VANE ANEMOMETER	ALNOR	RVA801	A05381	3/6/2023
☒ AMPROBE	FLUKE	325	40020345WS	3/6/2023
☐ TACHOMETER	MITUTOYO	PH-100-A	A16A95019P	3/6/2023
☐ HYDRONIC MANOMETER	TSI	HM-685	72009035	3/13/2023
☐ SOUND METER	QUEST TECH.	2900 SLM	CDC060026	4/4/2023
☐ VIBRATION	BALMAC INC.	200	0811096	3/29/2023
☐ POUND GAUGE	DWYER DIGITAL	DPG-005	4366516	3/6/2023
☐ HUMIDITY & TEMPERATURE	FLUKE #2	971	38401145	3/6/2023
☐ ULTRASONIC WATER FLOW METER	FUJI ELECTRIC	FSCS	N0F0844T	3/10/2023

FAN TEST SHEET

Job Number: 20531B

Date: 11/17/2023
System: EF-1

UNIT INFORMATION			
Unit Number	EF-1	Unit Manufacturer	GREENHECK
Location	ROOF	Model Number	FJI-20-BI-41-4-100-I-A50-X
Area Served	2nd FLOOR LAB	Serial Number	16479485 20D

FAN DESIGN / TESTED DATA					
	DESIGN	TESTED	SHEAVE AND BELT DATA		
Sched./Submit CFM	-	Traverse -	Motor Dia.	-	
Total Outlet/Inlet CFM	-	6,598	Motor adj. PD	-	
Total Return Air CFM	-	-	Motor Fixed PD	-	
Total Outside Air CFM	-	-	Motor Shaft	-	
Fan RPM	-	-	Fan Dia.	-	
TSP / ESP	- / -	- / 1.63	Fan Shaft	-	
TSP/ESP Suction Static Pressure		- / 1.52	Belts Size/Qty.	-	
TSP/ESP Discharge Static Pressure		- / 0.11	Shaft C to C	-	
Filter Pressure Drop		-	Direct Drive	YES	
			SCP	(1)	

MOTOR DATA						
NAMEPLATE				ACTUAL		
Motor Manufacturer	BALDOR	FLA	6.60	5.70	5.80	5.70
Motor HP	5.00	Voltage	460	470	470	469
Motor Phase	3	RPM	1,750	-		
Motor SF	1.15	BHP	-	4.34		
Motor Frame	184T	Speed SP		60 HZ		

Remarks:

(1) Speed control was done by scaling all of the fume hood from 25 HZ to 60 HZ.



LABORATORY HOOD TEST

Job Number: 20531B
Hood #: 1

Date: 11/17/2023
Room #: 2nd FLOOR LAB

RECORDED VELOCITY READING

153	137	142
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SIZE OF HOOD OPENING	12.5 "	x	39 "	=	3.39	SQ.FT.
LOWEST VELOCITY	137					
AVERAGE VELOCITY	144	x	3.39	SQ.FT. =	488	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	5/4/2023	
TECHNICIAN	MICHAEL CONNOLLY	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	YES					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:



LABORATORY HOOD TEST

Job Number: 20531B
Hood #: 2

Date: 11/17/2023
Room #: 2nd FLOOR LAB

RECORDED VELOCITY READING

104	115	133
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SIZE OF HOOD OPENING	12.5 "	x	39 "	=	3.39	SQ.FT.
LOWEST VELOCITY	104					
AVERAGE VELOCITY	117	x	3.39	SQ.FT. =	397	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	5/4/2023	
TECHNICIAN	MICHAEL CONNOLLY	TBE		MATT BRENNAN		
SASH MARKER INSTALLED				YES		
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:



LABORATORY HOOD TEST

Job Number: 20531B
Hood #: 3

Date: 11/17/2023
Room #: 2nd FLOOR LAB

RECORDED VELOCITY READING

107	106	108
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SIZE OF HOOD OPENING	12.5 "	x	39 "	=	3.39	SQ.FT.
LOWEST VELOCITY	106					
AVERAGE VELOCITY	107	x	3.39	SQ.FT. =	362	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	5/4/2023	
TECHNICIAN	MICHAEL CONNOLLY	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	YES					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:

Remarks:

Remarks:



LABORATORY HOOD TEST

Job Number: 20531B
Hood #: 6

Date: 11/17/2023
Room #: 2nd FLOOR LAB

RECORDED VELOCITY READING

126	123	121
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SIZE OF HOOD OPENING	12.5 "	x	39 "	=	3.39	SQ.FT.
LOWEST VELOCITY	121					
AVERAGE VELOCITY	123	x	3.39	SQ.FT. =	418	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	5/4/2023	
TECHNICIAN	MICHAEL CONNOLLY	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	YES					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:



LABORATORY HOOD TEST

Job Number: 20531B
Hood #: 7

Date: 11/17/2023
Room #: 2nd FLOOR LAB

RECORDED VELOCITY READING

114	108	100	117	102	97	99
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SIZE OF HOOD OPENING	12.5 "	x	87 "	=	7.55	SQ.FT.
LOWEST VELOCITY	97					
AVERAGE VELOCITY	105	x	7.55	SQ.FT. =	795	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	5/4/2023	
TECHNICIAN	MICHAEL CONNOLLY	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	YES					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:



LABORATORY HOOD TEST

Job Number: 20531B
Hood #: 8

Date: 11/17/2023
Room #: 2nd FLOOR LAB

RECORDED VELOCITY READING

126	116	100	89	121	115	108
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SIZE OF HOOD OPENING	12.5 "	x	87 "	=	7.55	SQ.FT.
LOWEST VELOCITY	89					
AVERAGE VELOCITY	111	x	7.55	SQ.FT. =	836	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	5/4/2023	
TECHNICIAN	MICHAEL CONNOLLY	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	YES					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:



LABORATORY HOOD TEST

Job Number: 20531B
Hood #:

Date: 11/17/2023
Room #: 2nd FLOOR LAB

RECORDED VELOCITY READING

106	102	99	121	119	114	93
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SIZE OF HOOD OPENING	12.5 "	x	87 "	=	7.55	SQ.FT.
LOWEST VELOCITY	93					
AVERAGE VELOCITY	108	x	7.55	SQ.FT. =	813	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	5/4/2023	
TECHNICIAN	MICHAEL CONNOLLY	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	YES					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:

DIFFUSER AND GRILLE TEST SHEET

Job Number: 20531B

Date: 11/17/2023
System: SILQ ROOM

[illegible]

Remarks: Canopy hood on the first floor.

(1) Tested with all fume hoods opened on 2nd floor, and motor at 60 HZ.

(2) Tested with all fume hood's off, and only canopy on at 25 HZ. At both test needed a minimum of 500 FPM or more.